

Let σ denote the uniform measure of a unit $(d-1)$ -sphere and $f : \mathbb{R}^d \mapsto \mathbb{C}$. The spherical average operator, defined by

$$Af(x) = \int f(x-y)d\sigma(y),$$

is a well-studied object, with abundant variants and applications. Instead of the Lebesgue measure as our playground, we consider a Borel measure μ that has a compact support and satisfies the fractal condition

$$\mu(B(x,r)) \leq Cr^s,$$

where $B(x,r)$ refers to the ball centered at x of radius r , and $C > 0$ is a constant uniform in x and r for some fixed index $s \in (0,d)$. We will give sharp L^p - L^q estimates of A with respect to μ . Moreover, the generalized Radon transform and a bilinear spherical average will be discussed in this context.